CF+OYM
& RM

Based on
ARBTS-13 to 15

Aakash
Medical | IIT-JEE | Foundations
(Divisions of Aakash Educational Services Limited)

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CONCEPT STRENGTHENING SHEET
CSS-05
Physics

ARBTS-13

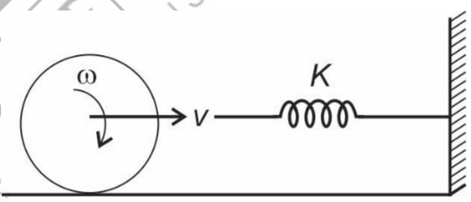
Q.3. (Mechanical wave)

1. A mechanical wave propagates in a medium along y axis. The particle of the medium
   (1) Must move along *x* axis
   (2) Must move along *Y* axis
   (3) May move along *Y* axis
   (4) Must move along *Z* axis
2. The speed of sound in medium depends on
   (1) Inertial property only
   (2) Elastic property only
   (3) Both elastic and inertial property
   (4) None
3. When a plane progressive harmonic mechanical wave, propagates in a non-dissipative medium then
   (1) Amplitude of all particles are same
   (2) Amplitude of all particles are different
   (3) Amplitude of all particle may be different
   (4) Both (2) & (3)
4. Particles of a plane progressive wave propagating in a non-dissipative medium executes
   (1) S.H.M.
   (2) Non periodic motion
   (3) Circular motion
   (4) Oscillatory but not SHM
5. The velocity of a plane progressive wave, propagating in a non-dissipative medium
   (1) Does not depend on nature of medium
   (2) Depends on the nature of medium
   (3) Directly proportional to elasticity of medium
   (4) All of these

ARBTS-13

Q.8 (Energy Conservation)

1. A hollow sphere of mass m is rolling with a speed *V* on a smooth horizontal surface and strikes a massless spring of force constant *K* attached to a massless smooth platform as shown in the figure. Then at time of maximum compression in spring



   (1) Total kinetic energy of sphere will be stored in spring as potential energy
   (2) Only Rotational K.E. of sphere will be stored in spring as potential energy
   (3) Only translational K.E. will be stored in spring a potential energy
   (4) Spring will not be compressed
2. In above question, the angular velocity of hollow sphere
   (1) Increases (2) Decreases
   (3) Remains same (4) Data is insufficient
3. In (Q.-1) When sphere strikes the spring, then torque on sphere, due to spring force will
   (1) Increase with increase in compression *X*
   (2) Decrease with increase in compression *X*

(3) Remain same with increase in compression X and is non-zero

(4) Be zero

4. In Question (1), if the surface is rough, then

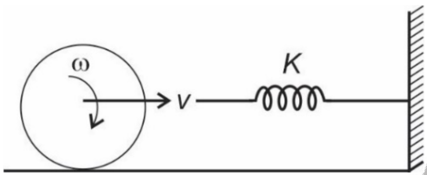
(1) Total kinetic energy of sphere will be stored in spring as potential energy

(2) Only Rotational K.E. of sphere will be stored in spring as potential energy

(3) Only translational K.E. will be stored in spring a potential energy

(4) Spring will not be compressed

5. A hollow sphere of mass m is rolling with a speed V on a rough horizontal surface and strikes a massless spring of force constant K attached to a massless smooth platform as shown in the figure. Then the maximum compression in spring is



(1) $\left(\sqrt{\dfrac{m}{k}}\right)v$ (2) $\left(\sqrt{\dfrac{2m}{3k}}\right)v$

(3) $\left(\sqrt{\dfrac{m}{5k}}\right)v$ (4) $\left(\sqrt{\dfrac{5m}{3k}}\right)v$

ARBTS-14

Q.36. (Magnetic Moment of electron in any orbit)

1. Magnetic moment of electron in any orbit of radius r, orbiting with speed v is

(1) $\dfrac{evr}{2}$ (2) evr

(3) $\dfrac{evr}{4}$ (4) $2evr$

2. An electron is orbiting around a nucleus with speed v. Magnetic moment of electron is proportional to

(1) v (2) v^{-3}

(2) v^{-2} (4) v^{-1}

3. An electron is orbiting around a nucleus in an orbit with principal quantum number ‘n’, then magnetic moment of electron is proportional to

(1) n^2 (2) n

(3) n° (4) n^{-1}

4. An electron revolve around the nucleus. The angular momentum of electron in that orbit is L. Then magnetic moment of electron is proportional to

(1) L^2 (2) L^{-2}

(3) L (4) L°

5. An electron of charge e moves in a circular orbit of radius r around the nucleus at a frequency ν. The magnetic moment associated with the orbital motion of electron is

(1) $\pi \nu e r^2$ (2) $\dfrac{\pi e r^2}{\nu}$

(3) $\dfrac{\pi \nu r^2}{e}$ (4) $\dfrac{\pi \nu e}{r}$

ARBTS-14

Q.43. (Value of electric field in electromagnetic wave)

1. The peak value of electric field at a distance r from the point source of electromagnetic wave of power P is

(1) $\sqrt{\dfrac{p}{\varepsilon_0 c\pi r^2}}$ (2) $\sqrt{\dfrac{P}{2\varepsilon_0 c\pi r^2}}$

(3) $\sqrt{\dfrac{P}{4\varepsilon_0 c\pi r^2}}$ (4) $\sqrt{\dfrac{p}{8\varepsilon_0 \pi r^2 c}}$

2. The amplitude of electric field in a parallel light beam of intensity 4 w/m^2 is

(1) 65.5 N/C (2) 55.5 N/C

(3) 45.5 N/C (2) 35.5 N/C

3. The peak value of electric field at a distance r from the point source of electromagnetic wave of power P is

(1) $\sqrt{\dfrac{\mu_0 CP}{2\pi r^2}}$ (2) $\sqrt{\dfrac{\mu_0 CP}{4\pi r^2}}$

(3) $\sqrt{\dfrac{\mu_0 CP}{\pi r^2}}$ (4) $\sqrt{\dfrac{2\mu_0 CP}{\pi r^2}}$

4. The intensity of electromagnetic wave is

(1) $\dfrac{E_o B_o}{2\mu_o}$ (2) $\dfrac{E_o^2}{2\mu_o}$

(3) $\dfrac{B_o^2}{2\mu_o}$ (4) All of these

5. The intensity of electromagnetic wave is

(1) $\dfrac{E_0^2}{2\mu_o}$ (2) $\dfrac{B_o^2 C}{2\mu_o}$

(3) $\dfrac{E_o^2}{2\mu_o C}$ (4) Both (2) and (3)

ARBTS-15

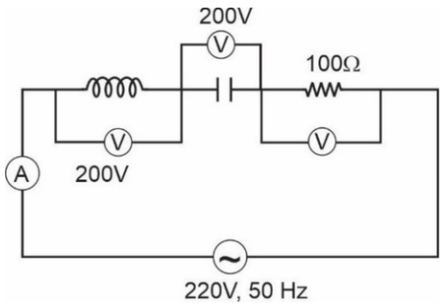
Q.12. Resonant Circuits

1. In an ac circuit containing an inductance and a capacitor in series. The current is found to be maximum when the value of inductance is 0.5 H and capacitance is 8 μF. The angular frequency of input ac must be

(1) 100 π rad/s (2) 200 π rad/s

(3) 500 rad/s (4) 500 π rad/s

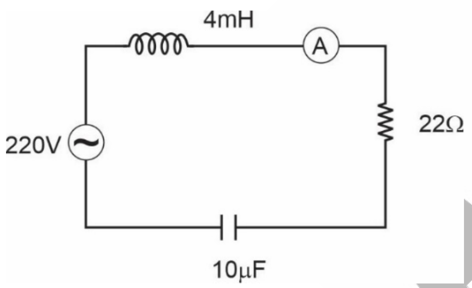
2. The reading of ammeter and voltmeter in the following circuit are respectively



(1) 2A, 200 V (2) 1.5 A, 100 V

(3) 1.7 A, 200 V (4) 2.2 A, 220 V

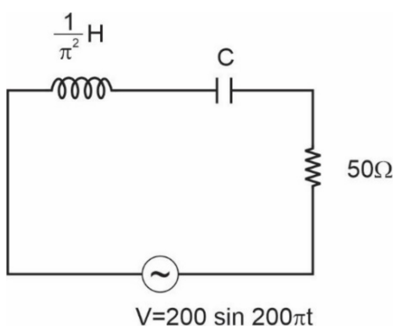
3. For the LCR circuit shown in figure. The resonance frequency and reading of ammeter at resonance frequency is



(1) 5000 Hz, 10 A (2) 5000 Hz, $10\sqrt{2}$ A

(3) 5000 rad/s, 10 A (4) 5000 rad/s, $10\sqrt{2}$ A

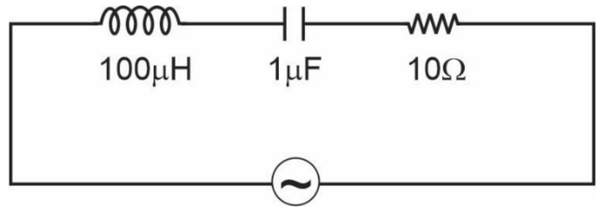
4. In the circuit shown in figure the value of capacitance C, so the current in circuit is maximum is



(1) 25μF (2) 20μF

(3) 50μF (4) 22.5μF

5. The following series LCR circuit, when driven by source of angular frequency 100 kilo-radian per second, the circuit effectively behave like



(1) Purely resistive

(2) Series R-L circuit

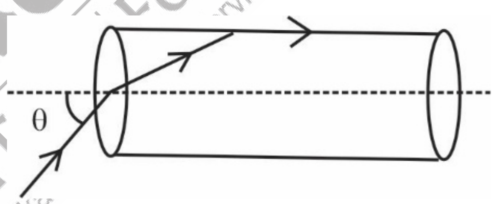
(3) Series R-C circuit

(4) Series RLC circuit

ARBTS-15

Q.21. (Total internal reflection)

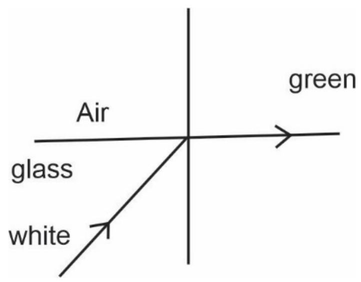
1. A transparent solid cylindrical rod has refractive index $\frac{2}{\sqrt{3}}$. It is surrounded by air. A light ray is incident at the mid point of one end of rod as shown in figure. The incident angle θ for which the light ray grazes along the wall is



(1) $\sin^{-1}\left(\frac{1}{2}\right)$ (2) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(3) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (4) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

2. When light is incident on the interface of glass and air as shown in figure. If green light is just totally internally reflected then the emerging ray in air contains,



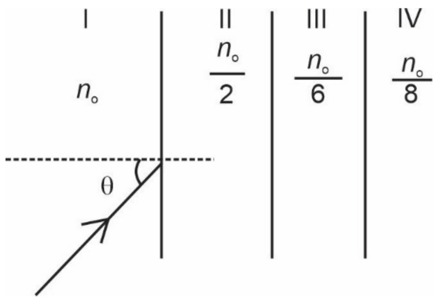
(1) Yellow, orange, red

(2) Violet, indigo, blue

(3) All colours

(4) All colours except green

3. A light beam is travelling from region I to region IV refer figure. The refractive index in regions I, II, III and IV are respectively, n_o, $\frac{n_o}{2}$, $\frac{n_o}{2}, \frac{n_o}{6}, \frac{n_o}{8}$ respectively. The angle of incidence θ for which the beam just misses entering region IV is



(1) $\text{Sin}^{-1}\left(\frac{3}{4}\right)$ (2) $\sin^{-1}\left(\frac{1}{8}\right)$

(3) $\sin^{-1}\left(\frac{1}{4}\right)$ (4) $\text{Sin}^{-1}\left(\frac{1}{3}\right)$

4. The wavelength of light in two liquids A and B is 3500 Å and 7000 Å. Then the critical angle of A relative to B will be.

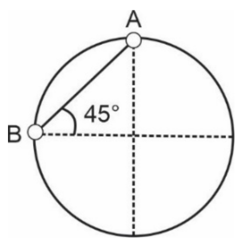
(1) 60° (2) 45°

(3) 30° (4) 15°

ARBTS-14 (10,16,27,30)

Q. 10 (Application Newtons second law)

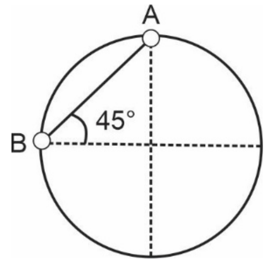
1. Two objects *A* and *B* each of mass m = 1kg are connected by a light inextensible string. They are restricted to move on a frictionless ring of radius *R* in a vertical plane (as shown in fig). The object are released from rest at position shown. Then, the tension in the string just after released is nearly (g = 10 m/s^2)



(1) 5 (2) $5\sqrt{2}$ N

(3) 10 N (4) $10\sqrt{2}$ N

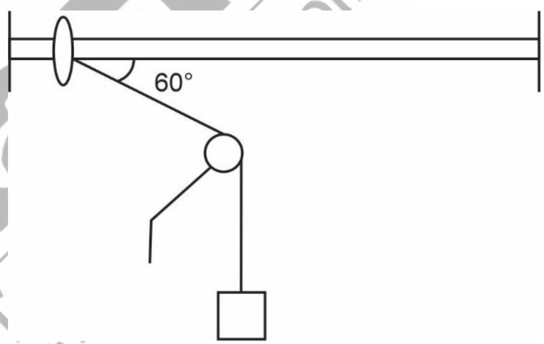
2. Two objectss *A* and *B* each of mass m are connected by a light inextensible string. They are restricted to move on frictionless ring of radius *R* in a vertical plane (as shown in fig). The objects are released from rest at the position shown. Then, the acceleration of mass B just offer released is nearly



(1) 10 m/s^2 (2) $5\sqrt{2}$ m/s^2

(3) 5 m/s^2 (4) $10\sqrt{2}$ m/s^2

3. A ring and a block each having mass m are connected by a light inextensible string which is passing over the pulley. The objects are released from rest at this position as shown in figure. The acceleration of the ring just after the release will be



(1) $\frac{2g}{5}$ (2) $\frac{4g}{5}$

(3) $\frac{3g}{5}$ (4) $\frac{g}{2}$

4. In the above question acceleration of the black will be

(1) $\frac{g}{5}$ (2) $\frac{2g}{5}$

(3) $\frac{3g}{5}$ (4) $\frac{g}{4}$

5. In question number (3) tension in the string will be

(1) $\frac{mg}{5}$ (2) $\frac{2mg}{5}$

(3) $\frac{4mg}{5}$ (4) $\frac{3mg}{5}$

ARBTS-14

Q.16 (Moment of inertia)

1. Consider the following statements

   (a) Perpendicular axis theorem for moment of inertia would be applicable for for any type of object

   (b) Parallel axis theorem applicable only for planner object

   The correct statement is/are

   (1) Only a (2) Only b

   (3) Both a and b (4) Neither a nor b

2. Consider the following two statements

   (a) among a given set of axis for a given body the moment of inertia is always maximum about one passing through the centre of mass

   (b) Perpendicular axis theorem for moment of would be applicable for a disc.

   The incorrect statement is/are

   (1) Only a (2) Only b

   (3) both a and b (4) Neither a nor b

3. Consider the followings statements

   (a) Perpendicular axis theorem is applicable for only planer object like lamina.

   (b) Parallel axis theorem is applicable any type of object

   The correct statement is/are

   (1) Only a (2) Only b

   (3) Both a and b (4) neither a nor b

4. Consider the following statements

   (a) The moment of inertia of solid sphere of mass M and radius R about an axis passing through its centre of mass is $\frac{2}{5}MR^2$

   (b) Parallel axis theorem for moment of inertia will be applicable for solid sphere

   The correct statement (S) is/are

   (1) Only a (2) Only b

   (3) Both a and b (4) Neither a nor b

5. Consider the followings

   (a) The moment of inertia of a ring of mass m and radius R about its diametrical axis is $\frac{mR^2}{2}$

   (b) Parallel axis theorem for moment of inertia would be applicable for ring

   The correct statement (s) is/are

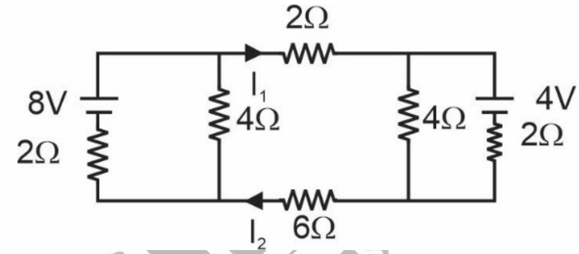
   (1) Only a (2) Only b

   (3) both a and b (4) neither a nor b

ARBTS-14

Q.27. (K.V.L.)

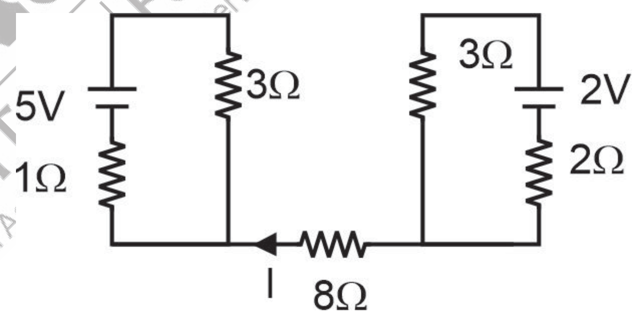
1. In the circuit shown the ratio of $I_1 : I_2$ is equal to



   (1) 1 : 1 (2) 1 : 2

   (3) 4 : 3 (4) 3 : 5

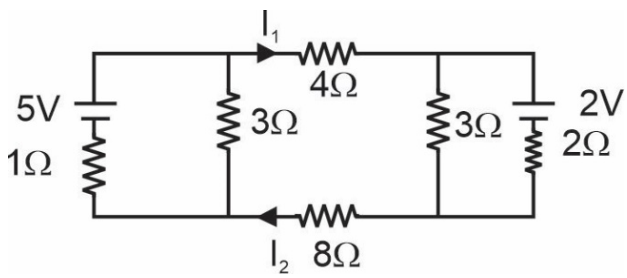
2. In the circuit shown the value of current I is equal to



   (1) 2A (2) 4A

   (3) 1A (4) Zero

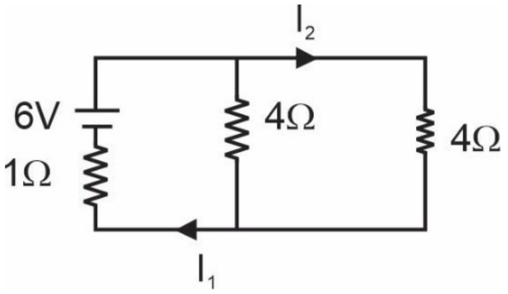
3. In the circuit shown the ratio of $I_1 : I_2$ when both 3Ω resistance remove from the circuit will



   (1) Increase (2) Decrease

   (3) Remains same (4) Cant say

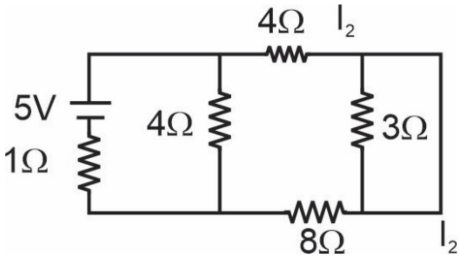
4. In the circuit shown, the ratio $I_1 : I_2$ is equal to



(1) 2 : 1 (2) 1 : 1

(3) 4 : 1 (4) 3 : 4

5. In the shown the current flowing through the 3Ω resistance will be



(1) $\frac{3}{2}$ A (2) 2A

(3) 1 A (4) Zero

ARBTS-14

Q. 30 (Motional e.m.f.)

1. A metallic rod of length L = 2 m and mass M = 2 kg is moving under the action of two unequal forces F_1 = 10 N and F_2 = 20 N (directed opposite to each other, acting at its end along its length. Ignore gravity and any external magnetic field. If specific charge of electron is α C/kg, then magnitude of electric field strength inside the rod in steady state would be

(1) $\frac{10}{\alpha}$ N/C (2) $\frac{5}{\alpha}$ N/C

(3) $\frac{20}{\alpha}$ N/C (4) $\frac{15}{\alpha}$ N/C

2. A metallic rod of length 3m and mass 2 kg is moving under the action of two unequal force F_1 = 10 N and F_2 = 30 N (directed opposite to each other) acting at its ends along it length ignore gravity and any external magnetic field. If specific charge of election is α C/kg. Then potential difference between the end of the rod in steady state will be

(1) $\frac{10}{\alpha}$V (2) $\frac{20}{\alpha}$V

(3) $\frac{30}{\alpha}$V (4) $\frac{5}{\alpha}$V

3. A metallic rod of length 3 m and 4 kg is moving under the action of two unequal forces F_1 = 20 N and F_2 = 30N (directed opposite to each other acting at its centre of mass perpendicular to its length. Ignore gravity and any external magnetic field. If specific charge of electron is α C/kg then the potential difference between the ends of the rod in steady state must be C

(1) $\frac{10}{\alpha}$V (2) $\frac{20}{\alpha}$V

(3) $\frac{30}{\alpha}$V (4) Zero

4. A metallic rod of length L = 3 m and mass 4 kg is moving under the action of two under the action of two unequal force F_1 = 10 N and F_2 = 30 N (directed opposite to each other) acting at its ends along its length. Ignore gravity and any external magnetic field. If specific charge of electrons is α C/kg Then magnitude of electric field inside the would be

(1) $\frac{10}{\alpha}$ N/C (2) $\frac{20}{\alpha}$ N/C

(3) $\frac{5}{\alpha}$ N/C (4) Zero

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CSS-05

Physics

ANSWERS

Q.3. (Mechanical Wave)

1. (3)
2. (3)
3. (1)
4. (1)
5. (2)

Q.8. (Energy conservation)

1. (3)
2. (3)
3. (4)
4. (1)
5. (4)

Q.36. (Magnetic Moment of electron in any orbit)

1. (1)
2. (4)
3. (2)
4. (3)
5. (1)

Q.43. (Value of electric field in electromagnetic wave)

1. (2)
2. (2)
3. (1)
4. (1)
5. (4)

Q.12. (Resonant circuits)

1. (3)
2. (4)
3. (3)
4. (1)
5. (1)

Q. 21. (Total Internal reflection)

1. (4)
2. (1)
3. (2)
4. (3)

Q. 10 (Application Newtons second law)

1. (2)
2. (3)
3. (1)
4. (1)
5. (3)

Q.16 (Moment of inertia)

1. (4)
2. (1)
3. (3)
4. (3)
5. (3)

Q.27. (K.V.L.)

1. (1)
2. (4)
3. (3)
4. (1)
5. (4)

Q. 30 (Motional e.m.f.)

1. (2)
2. (3)
3. (4)
4. (3)

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